

Apex Inventory

UI Requirement Specification

Categories & Layout

Route: /categories

Document Version: 1.0.0

Project: ReactExample (Apex Store Inventory Management)

1. Overview

The Categories & Layout page manages product category definitions and their physical placement within the store. Each category maps to a specific aisle, section, and shelf location. A visual aisle layout map shows how categories are distributed across the store.

2. Purpose

- Define and organize product categories for inventory classification.
- Map categories to physical store coordinates (aisle, section, shelf).
- Provide a visual store layout overview for staff orientation.
- Enable category deletion with dependency warnings.

3. Layout

- Top row: Create Category form (left) + Configured Categories list (right)
- Bottom: Visual Aisle Layout map (full width)

4. Create Category Form

4.1 Category Name

Required text input. Placeholder: 'e.g. Fresh Produce, Canned, Snacks'.

4.2 Description

Optional textarea (3 rows) for category description.

4.3 Store Placement Coordinates

- Aisle: Aisle 1-5 with descriptive labels (Dairy & Drinks, Bakery & Snacks, etc.)
- Section: Section A-D (Front-end, Middle, Back-end, Promo Endcap)
- Shelf Level: Shelf 1-4 (Bottom to Top/Eye Level)

4.4 Submit

Button: 'Create & Map Category'. POST /api/categories. Resets form to defaults on success.

5. Configured Categories List

Scrollable list (max-height 420px) showing all categories. Each card displays:

- Category name (title)
- Description or 'No description provided.'
- Badges: aisle, section, shelf
- Delete button (trash icon)

Empty state: 'No categories registered' with guidance to create first category.

6. Delete Category Behavior

Before deletion, checks if any inventory items use this category (case-insensitive name match).

- If items exist: warning confirm with item count and unmapped category notice.
- If no items: standard confirm dialog.
- On confirm: DELETE /api/categories/:id, refresh list.

Note: Deleting category does NOT remove or reassign inventory items.

7. Visual Aisle Layout

Grid of 5 aisle cells (Aisle 1 through Aisle 5). Each cell shows:

- Aisle label at top
- Category names assigned to that aisle (or 'Empty Aisle' if none)
- Active styling when aisle has at least one category
- Tooltip on category: name, section, shelf

Empty state when no categories: 'No Aisle Maps Available'.

8. Aisle Grouping Logic

Categories grouped by aisle field. Categories without aisle map to 'Unmapped' key (not shown in the 5-aisle grid which only displays Aisle 1-5).

9. Data Dependencies

- GET /api/categories - list and refresh
- GET /api/items - dependency check on delete
- POST /api/categories - create
- DELETE /api/categories/:id - remove

10. Technical Notes

- Component: src/pages/Categories.tsx